

JOEL VIVEK

Data Science • Machine Learning • Artificial Intelligence

codecatalyst001@gmail.com | github.com/CodeCatalyst001 | Andhra Pradesh, India

PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate (B.Tech, 3rd Year) with hands-on experience building end-to-end machine learning pipelines and data-driven predictive models. Proficient in Python, Scikit-learn, TensorFlow, and SQL, with demonstrated expertise in supervised and unsupervised learning, feature engineering, and exploratory data analysis (EDA). Built and deployed AI-powered systems spanning fault detection, credit risk, insurance fraud, and healthcare — seeking a Data Science / Machine Learning Engineer internship to deliver measurable business impact.

TECHNICAL SKILLS

Languages	Python, SQL, JAVA, C, JavaScript
ML / AI	Scikit-learn, TensorFlow, TensorFlow.js, OpenAI API, Linear Regression, Logistic Regression, Random Forest, Gradient Boosting, XGBoost, Neural Networks, NLP, Supervised & Unsupervised Learning, Classification, Regression, Fault Detection, Anomaly Detection, Fraud Detection
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Data Cleaning, EDA, Predictive Modeling, Statistical Analysis
Databases	MySQL, SQLite, MongoDB Atlas, Firebase Firestore
Frameworks	React.js, Node.js, Express.js, REST APIs, JWT Auth
Tools & Platforms	Git, GitHub, Jupyter Notebook, VS Code, Vercel, Render, Figma
Blockchain	Ethereum, Solidity (basics), Smart Contracts, Blockchain Data Integrity, Decentralized Verification

PROJECTS

Fault Detection in Wafers Using Sensor Data

Python · Scikit-learn · Pandas · Random Forest · SVM · Anomaly Detection

- Developed a binary classification model to detect faulty semiconductor wafers from high-dimensional sensor readings (590+ features), directly impacting manufacturing quality control.
- Implemented automated data preprocessing pipelines including missing-value imputation, feature selection using PCA and variance thresholding, and SMOTE for class imbalance — improving F1-score by ~22% over baseline.
- Evaluated SVM, Random Forest, and KNN classifiers; deployed best model achieving 96% accuracy and 0.94 F1-score on hold-out test set, enabling near-zero defective-unit shipments.
- Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook

Credit Card Defaulter Prediction

Python · Scikit-learn · XGBoost · Logistic Regression · Feature Engineering

- Built a binary classification model to predict credit card payment default risk using demographic and behavioral payment history data of 30,000+ customers.
- Applied SMOTE oversampling, RFE feature selection, and hyperparameter tuning (GridSearchCV) on Logistic Regression, Random Forest, and XGBoost; XGBoost achieved best AUC-ROC of 0.91.
- Engineered 8 derived financial features (payment ratio, credit utilization trends) that improved model recall on minority default class by 15%, critical for minimizing financial exposure.
- Technologies: Python, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, Jupyter Notebook

Insurance Fraud Detection

Python · Scikit-learn · Gradient Boosting · NLP · Anomaly Detection

- Designed a fraud detection pipeline for insurance claim data, combining structured tabular features with NLP-processed claim descriptions to identify high-risk fraud patterns.
- Applied TF-IDF vectorization on unstructured claim text combined with numerical features; Gradient Boosting classifier achieved 93% precision and 89% recall on fraudulent claims.
- Reduced false positive rate by 30% over rule-based baseline by integrating anomaly detection (Isolation Forest) as a pre-screening layer before classification.

- Technologies: Python, Pandas, Scikit-learn, NLTK, Gradient Boosting, Isolation Forest, Matplotlib

Cement Strength Prediction

Python · Pandas · Scikit-learn · Regression · Feature Engineering

- Built a regression model to predict compressive strength of concrete/cement mixtures from 8 material composition features, supporting civil engineering quality assurance workflows.
- Compared Linear Regression, Ridge, Lasso, and Gradient Boosting Regressor; Gradient Boosting delivered best R^2 of 0.92 and RMSE of 4.1 MPa on the test set.
- Performed correlation analysis and interaction feature engineering (water-cement ratio, aggregate ratios) that reduced prediction error by ~20% over the unengineered baseline.
- Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

House Price Prediction Model

Python · Pandas · Scikit-learn · NumPy · Matplotlib · Seaborn

- Engineered a regression-based predictive model to estimate residential property prices from 15+ features (location, sq. footage, amenities) using a dataset of 10,000+ records.
- Evaluated Linear Regression, Random Forest, and Gradient Boosting; achieved R^2 of 0.89 on test set; reduced prediction RMSE by ~18% over baseline via outlier detection and label encoding.
- Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

University Admission Prediction System

Python · Pandas · Scikit-learn · Logistic Regression · Random Forest

- Built a multi-class classification model to forecast graduate admission probability from GRE score, TOEFL, CGPA, LOR/SOP strength, and research experience.
- Random Forest delivered best accuracy of 92%; identified CGPA and GRE as top predictors via feature importance analysis, enabling a leaner model with no accuracy loss.
- Technologies: Python, Pandas, Scikit-learn, NumPy, Matplotlib, Jupyter Notebook

MediConnect — Unified AI + Blockchain Healthcare Platform

React.js · Node.js · MongoDB · TensorFlow.js · Ethereum · JWT

- Led technology & pitch for a full-stack telemedicine platform integrating AI diagnostics, blockchain medical records, and real-time video consultation (Jitsi), targeting rural healthcare gaps in India.
- Implemented an AI Health Assistant chatbot for symptom self-assessment using TensorFlow.js and OpenAI API; built Ethereum/Solidity smart contracts for tamper-proof patient record storage.
- Delivered end-to-end working prototype in a 12-hour hackathon sprint, covering ML integration, blockchain verification, and business pitch. [Team NULL POINTERS, 2025]
- Technologies: React.js, Node.js, MongoDB Atlas, TensorFlow.js, OpenAI API, Ethereum, Solidity, JWT, Figma

HACKATHON & COMPETITION EXPERIENCE

Technology & Presentation Lead — MediConnect Hackathon

Team NULL POINTERS, 2025

- Delivered end-to-end solution from ideation to working prototype in a 12-hour sprint: architecture design, AI model integration, blockchain verification, and pitch deck.
- Presented technical solution to a judging panel, clearly communicating ML model selection, data-flow design, and the business viability of an AI + blockchain healthcare stack.
- Coordinated a cross-functional team (frontend, backend, design) under extreme time constraints, ensuring all deliverables were completed on schedule.

EDUCATION

Bachelor of Technology — Computer Science & Engineering

2023 – 2027 (Expected)

Andhra Pradesh, India

Expected Graduation: May 2027

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming (Python / C++)

Activities: Active participant in hackathons, tech clubs, and open-source projects; self-directed learning in blockchain integration and AI systems.

AREAS OF INTEREST

Machine Learning & AI Engineering • Data Science & Predictive Modeling • Fault Detection & Anomaly Detection • Fraud Detection & Risk Analytics • Natural Language Processing • Blockchain in Data Integrity • Healthcare Technology